

SOHAIL HARESH GIDWANI

Los Angeles, CA | *Open to relocation*

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EXPERIENCE

Keck School of Medicine, USC

Oct 2025 - Present

Student Worker / Research Assistant, Multi-Modal AI for Alzheimer's Disease

Los Angeles, CA

- Built **MEMOIR-VLM**, fusing T1 MRI, DTI, and clinical data across 2,363 ADNI subjects with cross-attention and modality dropout; achieved 91.3% balanced accuracy on CN vs dementia and 78.7% zero-shot on OASIS-3 (CN vs impaired, DTI absent); first-author paper published in *Frontiers in Computational Neuroscience* (2026). [Paper]
- Built **Neuro-Var-JEPA**, a self-supervised 3D Vision Transformer pipeline in PyTorch over 16,024 brain MRI scans from 4 cohorts; completed 82 pre-training runs and identified a data-regime crossover where variational JEPA outperformed the deterministic baseline on 3 tasks at 25% of pre-training data but underperformed at full data.
- Conducted the **Malliavin scaling-law uncertainty study**: 1,225 GPU training runs on ADNI measured uncertainty bands $7.1\text{--}14.4\times$ wider when encoders were trained from scratch on each subsample than when frozen; built resumable experiment drivers and compared adapter and partial fine-tuning controls to investigate the role of pretrained initialization.

Insaito, Inc.

May 2025 - Jul 2025

Senior Software Engineer - I

Remote

- Architected an AI agent platform in TypeScript and Python on serverless cloud infrastructure using LangChain and LangGraph for stateful, graph-based multi-step agent workflows across 100+ third-party integrations, with task routing, retries, and persistent context across long-running runs.
- Built event-driven backend services for concurrent agent orchestration with tool-calling, structured fallbacks, and a production tracing layer over agent outputs; owned core infrastructure for open-source LLM deployment, MCP servers, and prompt/evaluation harnesses for long tool-use chains.

IIFL Finance Ltd.

Jun 2023 - May 2025

Full Stack Software Developer

Mumbai, India

- Shipped production AI systems across two years at India's largest non-banking financial company, working on agent-driven customer interaction and workflow automation in a regulated financial services environment with integration into legacy core systems; embedded with the teams who would use each system before building.
- Designed CapitalGenie, a GPT-4o-powered support agent with custom prompt engineering and internal API orchestration that ingested unstructured user issue data, structured it, and generated personalized responses, accelerating resolution time by 70%.
- Built an internal RAG chatbot with a React frontend and a Python/Flask backend from scratch (Qdrant, Azure OpenAI), integrated into the Zoho ticketing flow and cutting support tickets by 20–25% in the first month. (Certificate of Achievement)
- Engineered an AI-powered Gold Loan Image Audit pipeline using GroundingDINO and Swin-Transformer for automated fraud detection on loan application photos, reducing potential loan fraud by 15%.

EDUCATION

University of Southern California

Aug 2025 - May 2027

M.S., Computer Science, GPA: 3.75/4.0

Los Angeles, CA

- **Coursework:** Applied NLP, Analysis of Algorithms, Information Retrieval & Web Search Engines, ML for Data Science

University of Mumbai

Aug 2019 - May 2023

B.E., Computer Engineering, CGPA: 9.05/10

Mumbai, India

SKILLS

AI/ML: PyTorch, TensorFlow, HuggingFace, scikit-learn, pandas, numpy, LangChain, LangGraph, Claude Code, fine-tuning, RAG, FAISS, cross-encoder reranking, agentic workflows, MCP, prompt engineering, LLM evaluation & benchmarking (Llama 3.2 3B, Qwen 2.5 7B, Mistral 7B, Gemma 4 26B MoE, MedGemma 1.5 4B), Vector DBs

Programming: Python, TypeScript, JavaScript, Java, C/C++, SQL

Backend & Cloud: FastAPI, Flask, Node.js, Express, Hono, RESTful APIs, AWS, Azure, GCP, Docker, Linux, Serverless, event-driven architectures

Databases & Tooling: PostgreSQL, pgvector, MySQL, MongoDB, Oracle, Qdrant, SQLAlchemy, Git, CI/CD, Test-Driven Development, Code Reviews

PROJECTS

- **Portage: Autonomous Flask→FastAPI Migration Agent:** Built a Postgres-checkpointed LangGraph agent that plans multi-file migrations, creates new modules, and verifies results in network-off Docker sandboxes. Shipped CLI/MCP tools and a Next.js dashboard for plans, diffs, and recovery evidence. Historical development gates reached 5/5 autonomous migrations each on Flaskr and Watchlist (24/24 and 15/15 tests); closed out with 357 passing backend tests. Kept held-out evaluation separate from development results. [GitHub]
- **CoT Faithfulness Analysis (USC NLP):** Designed a 4-experiment study of whether chain-of-thought causally drives LLM answers or is post-hoc rationalization. Ran $\sim 15\text{K}$ queries against Llama 3.2 3B and Qwen 2.5 7B on GSM8K and ARC-Challenge; found math CoT partially faithful (44–50pp drop under corruption) while science multiple-choice CoT is largely decorative. [link]
- **Knowledge Hub (RAG-Powered Q&A Agent):** Dockerized RAG agent on Flask and PostgreSQL+pgvector with hybrid full-text + vector retrieval, a citation-enforced Q&A pipeline via a local LLM, and multi-pass OCR for PDFs, images, and handwritten notes. [link]